



MATHEMATICS (EXTENSION 2)

2013 HSC Course Assessment Task 4

Monday August 12, 2013

General instructions

- Working time – 60 min
(plus 5 minutes reading time)
- Commence each new question on a new page.
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper and hand to examination supervisors.

Class (please ✓)

- ☐ 12M4A – Mr Choy
- ☐ 12M4B – Mr Weiss
- ☐ 12M4C – Ms Ziazaris

STUDENT NUMBER # BOOKLETS USED:

Marker's use only.

QUESTION	1	2	3	Total	%
MARKS	$\overline{15}$	$\overline{15}$	$\overline{13}$	$\overline{43}$	

Glossary

- $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3\}$ – set of all integers.
- $\mathbb{Z}^+$ – all positive integers (excludes zero)
- $\mathbb{R}$ – set of all real numbers

Question 1 (15 Marks) Commence a NEW page. **Marks**

- (a) Find the value of k in the expansion of $(1 + kx)^{10}$ if the coefficient of x^8 is 42 times the coefficient of x^6 in the expansion. **3**

- (b) Find the (numerically) greatest term in the expansion of $(2 + 5x)^{11}$ when $x = \frac{2}{3}$. **3**

- (c) i. For all $x \in \mathbb{R}$ and $n \in \mathbb{Z}^+$, show that **2**

$$\binom{n}{0}x^n + \binom{n}{1}x^{n-1}(1-x) + \binom{n}{2}x^{n-2}(1-x)^2 + \dots + \binom{n}{n}(1-x)^n = 1$$

- ii. Deduce that **1**

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n} = 2^n$$

- (d) Use the expansion of $(1 + x)^n$ to show that **3**

$$\sum_{r=0}^n \frac{\binom{n}{r}}{r+1} = \frac{2^{n+1} - 1}{n+1}$$

- (e) By considering the coefficients of x^{n+1} from both sides of the identity **3**

$$(1+x)^n(1+x)^n = (1+x)^{2n}$$

prove that

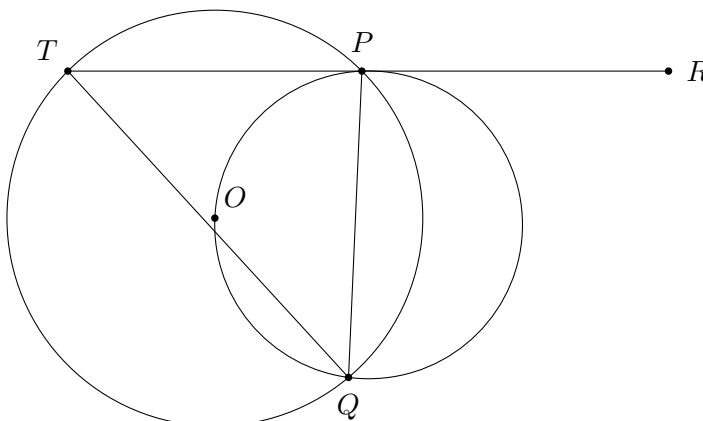
$$\binom{n}{0}\binom{n}{1} + \binom{n}{1}\binom{n}{2} + \dots + \binom{n}{n-1}\binom{n}{n} = \frac{2n!}{(n-1)!(n+1)!}$$

Question 2 (15 Marks)

Commence a NEW page.

Marks

- (a) Two circles intersect at P and Q as shown. The smaller circle passes through the centre, O , of the larger circle.



The tangent to the smaller circle RPT cuts the larger circle at T . PQ bisects $\angle RQO$. Let $\angle PTQ = \alpha$.

- i. Show that $\triangle PQT$ is isosceles. **3**
 - ii. Show that P is the midpoint of RT . **2**
- (b) If a, b, c and $d > 0$, show that
- i. $a^2 + b^2 + c^2 - ab - bc - ca \geq 0$ **2**
 - ii. $\frac{a+b+c}{3} \geq \sqrt[3]{abc}$ **2**
 - iii. $(a+b+d)(b+c+d)(c+a+d)(a+b+c) \geq 81abcd$ **2**
- (c) Consider the function $f(x) = (ax - b)^2 + (cx - d)^2$ where a, b, c and $d \in \mathbb{R}$.
- i. Explain why $f(x) > 0$ for all x . **1**
 - ii. Hence or otherwise, prove that **3**

$$|ab + cd| \leq \sqrt{a^2 + c^2} \sqrt{b^2 + d^2}$$

Examination continues overleaf...

Question 3 (13 Marks) Commence a NEW page. **Marks**

- (a) The Fibonacci sequence $1, 1, 2, 3, 5, 8, \dots$ can be defined as $u_1 = 1, u_2 = 1, u_{n+2} = u_n + u_{n+1}$ for integers $n > 0$.

i. Use induction to prove that $u_n < a^n$, for any $a > \frac{1 + \sqrt{5}}{2}$. **3**

ii. Assuming that $\frac{u_{n+1}}{u_n}$ approaches a limit as $n \rightarrow \infty$. Show that **4**

$$\frac{u_{n+1}}{u_n} \rightarrow \frac{1 + \sqrt{5}}{2}$$

(b) i. Write out the binomial expansion of $\left(1 + \frac{1}{n}\right)^n$, where $n \in \mathbb{Z}^+$. **1**

ii. Show that the $(k+1)$ -th term, T_{k+1} , is given by **2**

$$T_{k+1} = \frac{1}{k!} \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \cdots \left(1 - \frac{k-1}{n}\right)$$

iii. Let the $(k+1)$ -th term of the expansion $\left(1 + \frac{1}{n}\right)^{n+1}$ be U_{k+1} . **3**
Show that $U_{k+1} > T_{k+1}$.

End of paper.

$$Q1 \quad a) \quad (1+kx)^{10} = {}^{10}C_0 + {}^{10}C_1 kx + {}^{10}C_2 (kx)^2 + \dots + {}^{10}C_6 (kx)^6 + \dots + {}^{10}C_8 (kx)^8$$

$$\therefore 42 {}^{10}C_6 k^6 = {}^{10}C_8 k^8$$

$$42 \times \frac{10 \times 9 \times 8 \times 7}{4 \times 3 \times 2 \times 1} = \frac{10 \times 9}{2 \times 1} k^2$$

$$\frac{42 \times 10 \times 8 \times 7}{4 \times 3} = k^2$$

$$196 = k^2$$

$$\therefore k = \pm 14$$

$$b) \quad (2+5x)^{11}$$

$$\frac{T_{r+1}}{T_r} = \frac{{}^{11}C_r 2^{11-r} (5x)^r}{{}^{11}C_{r-1} 2^{12-r} (5x)^{r-1}}$$

$$= \frac{11!}{r!(11-r)!} \cdot \frac{5x}{2} \times \frac{(12-r)!(r-1)!}{11!}$$

$$= \frac{12-r}{r} \times \frac{5}{2} \times \frac{2}{3} > 1$$

$$60-5r > 3r$$

$$60 > 8r$$

$$r < 7\frac{3}{4}$$

$$r = 7$$

$$\therefore \text{greatest term} \quad \frac{11!}{4!7!} 2^4 5^7 \times \frac{2}{3}$$

$$c) \quad i) \quad (a+b)^n = {}^nC_0 a^n + {}^nC_1 a^{n-1} b + \dots + {}^nC_r a^{n-r} b^r + \dots + {}^nC_n b^n$$

$$\text{let } a = x \text{ and } b = 1-x$$

$$1 = {}^nC_0 x^n + {}^nC_1 x^{n-1} (1-x) + \dots + {}^nC_r x^{n-r} (1-x)^r + \dots + {}^nC_n (1-x)^n$$

$$ii) \quad \text{let } x = \frac{1}{2}$$

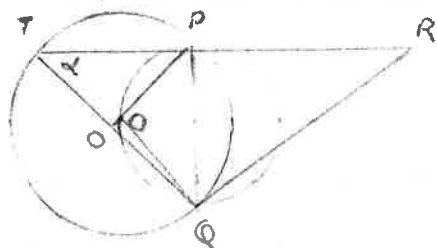
$$1 = {}^nC_0 \left(\frac{1}{2}\right)^n + {}^nC_1 \left(\frac{1}{2}\right)^{n-1} \left(\frac{1}{2}\right) + \dots + {}^nC_r \left(\frac{1}{2}\right)^{n-r} \left(\frac{1}{2}\right)^r + \dots + {}^nC_n \left(\frac{1}{2}\right)^n$$

$$1 = {}^nC_0 \frac{1}{2^n} + {}^nC_1 \frac{1}{2^n} + \dots + {}^nC_r \frac{1}{2^n} + \dots + {}^nC_n \frac{1}{2^n}$$

$$\therefore 2^n = {}^nC_0 + {}^nC_1 + \dots + {}^nC_r + \dots + {}^nC_n$$

Q2

a) i)



$$\angle POQ = 2 \angle PTQ = 2\alpha \quad (\text{angle at centre } 2\times \text{angle at circumference})$$

$$\angle P\hat{R} = \angle P\hat{O}Q = 2\alpha \quad (\text{alt. segment thm})$$

In $\triangle PTQ$

$$\hat{TOP} + \hat{PTQ} = \hat{QPR} \quad (\text{ext angle of } \Delta = \text{interior opposite})$$

$$\hat{T}Q P = P \hat{T}Q$$

$\therefore \Delta PTQ$ is isos (2 angles equal)

ii) ΔPOQ is isos. (radii equal)

$$\hat{OPQ} = \hat{OQP}$$

$$\therefore \hat{PQO} = \frac{1}{2} (180 - 2\alpha) \quad (\text{angle sum of } \triangle)$$

In ΔPQR

$$\angle PQR = \angle PQO = 90^\circ - \alpha \quad (\text{PQ bisects } \angle RQO)$$

$$\begin{aligned}\hat{P}RQ &= 180 - 2\alpha - (90 - \alpha) \quad (\text{angle sum of } \Delta) \\ &= 90 - \alpha\end{aligned}$$

$\therefore \triangle PRQ$ is isos (2 angles equal)

$$\therefore PR = PQ$$

But $PQ = PT$ from (i) uses Δ

$$\therefore TP = PR$$

and P is the midpoint of RT .

Q2)

i) The sum of 2 squares is positive.

$$\begin{aligned} \text{c) ii) } f(x) &= a^2x^2 - 2abx + b^2 + c^2x^2 - 2cdx + d^2 \\ &= (a^2 + c^2)x^2 - (2ab + 2cd)x + b^2 + d^2 \end{aligned}$$

Since $f(x) \geq 0$ for all x ,

the function has no real roots, i.e. $\Delta \leq 0$

$$\Delta = (- (2ab + 2cd))^2 - 4(a^2 + c^2)(b^2 + d^2) \leq 0$$

$$4(ab + cd)^2 - 4(a^2 + c^2)(b^2 + d^2) \leq 0$$

$$\begin{aligned} \frac{(ab + cd)^2}{\sqrt{(ab + cd)^2}} &\leq \frac{(a^2 + c^2)(b^2 + d^2)}{\sqrt{a^2 + c^2} \sqrt{b^2 + d^2}} \\ \sqrt{(ab + cd)^2} &\leq \sqrt{a^2 + c^2} \sqrt{b^2 + d^2} \end{aligned}$$

$$\text{i.e. } |ab + cd| \leq \sqrt{a^2 + c^2} \sqrt{b^2 + d^2} \quad \#$$

Q3

a)

ii) If $\frac{U_{n+1}}{U_n}$ approaches a limit as $n \rightarrow \infty$

then $\frac{U_{n+2}}{U_{n+1}}$ approaches the same limit.

$$\text{So as } n \rightarrow \infty \quad \frac{U_{n+2}}{U_{n+1}} = \frac{U_{n+1}}{U_n}$$

$$\text{i.e. } \frac{U_{n+1} + U_n}{U_{n+1}} = \frac{U_{n+1}}{U_n} \quad \text{since } U_{n+2} = U_n + U_{n+1}$$

$$\therefore 1 + \frac{U_n}{U_{n+1}} = \frac{U_{n+1}}{U_n}$$

$$\text{let } \frac{U_{n+1}}{U_n} = x \quad \text{as } n \rightarrow \infty$$

$$\therefore 1 + \frac{1}{x} = x$$

$$\text{i.e. } x^2 - x - 1 = 0$$

$$x = \frac{1 \pm \sqrt{5}}{2}$$

$$\text{but } x > 0 \quad \therefore \text{limit is } \frac{1 + \sqrt{5}}{2}$$

$$\text{i.e. as } n \rightarrow \infty, \quad \frac{U_{n+1}}{U_n} \rightarrow \frac{1 + \sqrt{5}}{2}$$